Sahaj Bhadja

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Summary

Software engineer pursuing an MSCS at USC with 2+ years building **full-stack applications**, **Java/Python-backed services**, **React interfaces**, and AI-enabled workflows across cloud systems, relational data, testing, and secure user-facing platforms.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Architected secure multi-tenant application services for 500+ users, reducing latency from **1.2s to 350ms** through SQL optimization, indexed access patterns, and backend improvements across cloud-backed workflows.
- Built automated validation and ETL checks for 5,000+ monthly jobs, reducing production failures and accelerating issue resolution through earlier detection of regressions, data-quality drift, and runtime anomalies.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Developed full-stack React and Flask features handling 10,000+ daily requests, translating product feedback into maintainable APIs and interfaces while improving usability, testing discipline, and delivery predictability.
- Built AI-assisted grading workflows with multimodal models, OCR, and relational persistence, reducing feedback turnaround by **60%** while supporting real-world validation by instructors in production environments.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built Python experimentation pipelines across 100,000+ trajectory records, enabling reproducible algorithm evaluation and reliable debugging while supporting rapid iteration on model-driven application behaviors.
- Produced technical reports and test scenarios translating findings into actionable improvements, strengthening cross-team communication, reducing handoff ambiguity, and improving engineering quality as prototypes matured toward deployment.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered API and ETL microservices using Python, AWS Glue, and S3, improving response time by **30%** while supporting distributed integrations, scalable data movement, and reliable release workflows.
- Maintained 20+ automated test suites using Jenkins, PyTest, and Mockito, increasing CI reliability by **68%**, reducing regressions, and improving confidence during frequent releases across coordinated services.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Developed MERN and WebSocket modules for 2M+ synchronized records across eight workflows, improving throughput consistency, reducing stale-session behavior, and strengthening real-time responsiveness for user-facing processes.
- Automated Docker-based build and test flows with Jenkins, cutting release time by **50%**, improving deployment consistency, and reducing environment drift that previously slowed validation and troubleshooting.

Skills

Languages: **Java**, **Python**, JavaScript/TypeScript, SQL, C++, Bash
Frontend: React, Next.js, Angular.js, responsive UI development, user-facing web workflows
Backend & APIs: Spring Boot, Flask, FastAPI, Node.js, REST APIs, microservices, ETL pipelines
Data & Cloud: PostgreSQL, MySQL, SQL Server, MongoDB, AWS, Azure, Docker, Kubernetes, Git/GitHub, Jenkins, CI/CD
AI & Tools: LLM applications, RAG concepts, LangChain familiarity, automated testing, Agile/Scrum

Projects

Big-Brain Live | *LLaMA-2, FAISS, PostgreSQL, GCP*

- Built a secure document platform with semantic search, summarization, and collaborative Q&A, enabling RAG-style workflows over 50K+ documents while reducing document-analysis time by **60%**.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Built an AI-powered SQL workspace supporting 10+ databases, accelerating query authoring and schema exploration through a unified **Java** and TypeScript codebase with Docker-backed development workflows.

PiXMod Live | *Next.js, MongoDB, Stripe, Transformers*

- Built an AI-powered Next.js SaaS platform for image editing and search, integrating Stripe billing and MongoDB-backed workflows while improving premium conversion by **18%** through polished experiences.